Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

# EIN 4360 Facilities Planning and Material Handling

## Chapter 5: Material Handling

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POWERING THE NEW ENGINEER TO TRANSFORM THE FUTURE

# Summary

- Learn about Material Handling and its impact on facilities planning
- Learn about the different Material Handling equipment

# Scope and definitions of Material Handling

In a typical industrial facility, material handling accounts for

~25% of all employees
~55% of all factory space
~87% of production time

Material Handling represents between 15% to 70% of the total cost of a manufactured product

# Scope and definitions of Material Handling

- **Material handling** is the art and science associated with the **movement, storage, control and protection of goods and materials** throughout the process of their manufacture, distribution, consumption, and disposal.

# Scope and definitions of Material Handling

- How can material handling affect the product?

Providing right **amount**:
- JIT (just in time)
- Coordinate issue quantities with stocking quantities

Providing right **material**:
- Part numbering system
- Accurate identification system (Bar codes)
- Simplifying the part numbering system

Providing right **condition**:
- Top quality– damage free
- Painted vs. unpainted?
- Packaged versus unpackaged

Providing right **sequence**:
- Productivity can be increased by eliminating unnecessary steps in an operation and improving those that remain.

# Scope and definitions of Material Handling

- How can material handling affect the product?

Providing right **orientation**:
- Positioning of materials for ease of handling

Providing right **place**:
- Transportation and storage
- Centralized vs. decentralized storage

Providing right **time**:
- On-time delivery (not early or late)
- Reduce variance in delivery times

Providing right **cost**:
- Most efficient system at most reasonable cost
- Not necessarily the lowest cost

Providing right **method**:
- Requirements driven material handling systems
- The right method will come from analysis of the other factors using a systems approach

# Material Handling Principles (10 principles from CIC-MHE)

- Planning Principle – MH should be the result of a **direct** and **intentional plan**
- Standardization Principle – MH should **minimize variety** and **customization** in the methods and equipment
- Work principle – MH should **NOT sacrifice productivity**
- Ergonomic Principle – MH should follow **ergonomic principles** (adapt the job to the worker)
- Unit load Principle – Units should be **sized appropriately** (consisting of individual items) so that they can be moved as one or several units

# Material Handling Principles (10 principles from CIC-MHE)

- Space utilization – Space is three dimensional, **don't forget to look up!**
- System principle – MH **considerations** must consider all of the interrelated or independent systems (e.g., receiving, inspection, shipping, etc.).
- Automation principle – **Automate** when possible (and fiscally sensible)
- Environmental principle – MH should seek to **minimize the environmental footprint**
- Life-cycle cost principle – Account for all economic considerations during the **life of MH equipment** (initial cost, maintenance, salvage, time, etc.)

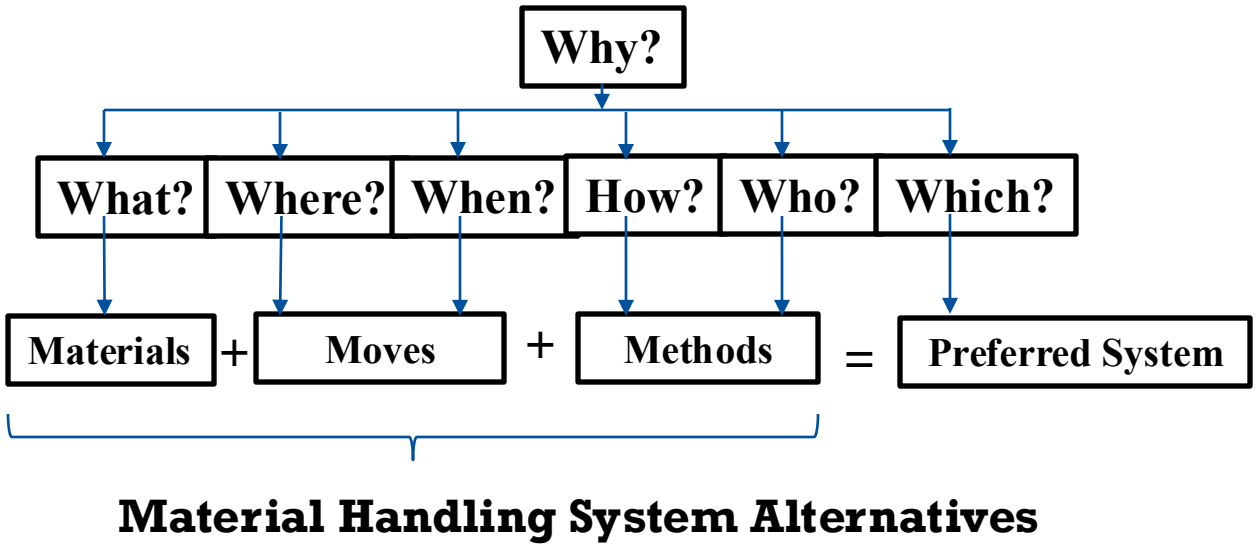
## Can you think any examples of these different principles in real world applications?

- Planning Principle
- Standardization Principle
- Work Principle
- Ergonomic Principle
- Unit Load Principle
- Space Utilization
- System Principle
- Automation Principle
- Environmental Principle
- Life-Cycle Cost Principle

Table 5.1 *Material Handling Audit Sheet*

| | | Condition Will Require: | | | | |
|---|---|---|---|---|---|---|
| Conditions Indicating Opportunities for Improvement | Condition Observed | Supervisor Attention | Management Attention | Analytical Study | Capital Investment | Other Comments |
| 1. Production equipment idle due to material shortage | | | | | | |
| 2. Material piled directly on floor | | | | | | |
| 3. In-plant containers not standardized | | | | | | |
| 4. Operators travel excessively for materials and supplies | | | | | | |
| 5. Excessive demurrage | | | | | | |
| 6. Misdirected material | | | | | | |
| 7. Backtracking of material | | | | | | |
| 8. Automatic data collection system not used | | | | | | |
| 9. Excessive trash removal | | | | | | |
| 10. System not capable of expansion and/or change | | | | | | |
| 11. No pre-kitting of work | | | | | | |
| 12. Crushed loads in block stacking | | | | | | |

# The Material Handling Systems Equation



Why might we only consider distance during the transport steps?

# Describing Flow

Date: 9-30-02
Analyst: TLR

Location: Graves Mountain
Process: Apple Sauce

A Flow Process Chart

| Step | Operation | Transport | Inspect | Delay | Storage | Description of process | Time (min) | Distance (feet) |
|---|---|---|---|---|---|---|---|---|
| 1 | ● | ⇨ | □ | D | ▽ | Unload apples from truck | 20 | |
| 2 | ○ | ➡ | □ | D | ▽ | Move to inspection station | | 100 ft |
| 3 | ○ | ⇨ | ■ | D | ▽ | Weigh, inspect, sort | 30 | |
| 4 | ○ | ➡ | □ | D | ▽ | Move to storage | | 50 ft |
| 5 | ○ | ⇨ | □ | D | ▼ | Wait until needed | 360 | |
| 6 | ○ | ➡ | □ | D | ▽ | Move to peeler | | 20 ft |
| 7 | ● | ⇨ | □ | D | ▽ | Apples peeled and cored | 15 | |
| 8 | ○ | ⇨ | □ | D | ▼ | Soak in water until needed | 20 | |
| 9 | ● | ⇨ | □ | D | ▽ | Place in conveyor | 5 | |
| 10 | ○ | ➡ | □ | D | ▽ | Move to mixing area | | 20 ft |
| 11 | ○ | ⇨ | ■ | D | ▽ | Weigh, inspect, sort | 30 | |
| | Page 1 0f 3 | | | | | Total | 480 | 190 ft |

# Designing Material Handling Systems

One way to organize data and generate alternatives is to use the *material handling planning chart*.

**O**: operation
**T**: transport
**S**: storage
**I**: inspection

MATERIAL HANDLING PLANNING CHART

Company A.R.C. Inc. Prepared by I.A. Layout Alternative 1
Product Air Speed Control Valve Date Sheet 1 of

| Step No. | O | T | S | I | Description | Oper No. | Dept. | Cont. Type | Size | Wt. | Qty. Per Cont. | Freq | Dist | Method of Handling |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | X | | Bar stock in storage (2200) | | Stores. | | | | | | | |
| 2 | | X | | | Profit stores to saw dept. | | | LDDSE (FK.TRK) | 2.5″ × 3.5 × 16″ | 5 lb | to bars | 3 times daily | 16 ft | Fork lift |
| 3 | | | X | | Store in saw department | | Saw | | | | | | | |
| 4 | X | | | | Cut to length | 0101 | Saw | | | | | | | |
| 5 | | X | | | From saw to grinding | | | TOTE pan | 15″ × 12″ × 7″ | 30 lb | 30 | Twice daily | 10 ft | Platform hand truck |
| 6 | | | X | | Store in grinding | | Grinding | | | | | | | |
| 7 | X | | | | Grind to length | 0201 | Grinding | | | | | | | |
| 8 | | | X | | | | | TOTE pan | 15″ × 12″ × 7″ | 30 lb | 30 | Twice daily | 13 ft | Platform hand truck |
| 9 | | | | X | Store in deburring | | Deburring | | | | | | | |
| 10 | X | | | | Deburr | 0301 | Deburring | | | | | | | |
| 11 | | X | | | From deburring to dr. Prs | | | TOTE pan | 15″ × 12″ × 7″ | 30 lb | 30 | Twice daily | 16 ft | Platform hand truck |
| 12 | | | X | | Store in drill press | | Drill Press | | | | | | | |
| 13 | X | | | | Dr. CD holes tap. rean, dsk | 0401 | Drill Press | | | | | | | |
| 14 | | X | | | From dr. press to tur. lathe | | | TOTE pan | 15″ × 12″ × 7″ | 30 lb | 30 | Twice daily | 33 ft | Platform hand truck |

**Where and When?**

# Designing Material Handling Systems

One way to organize data and generate alternatives is to use the *material handling planning chart*.

MATERIAL HANDLING PLANNING CHART

Company A.R.C. Inc, Prepared by I.A. Layout Alternative 1
Product Air Speed Control Value Date Sheet 1 of

| Step No. | O | T | S | I | Description | Oper No. | Dept. | Cont. Type | Size | Wt. | Qty. Per Cont. | Freq | Dist | Method of Handling |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | | | X | | Bar stock in storage (2200) | | Stores. | | | | | | | |
| 2 | | X | | | Profit stores to saw dept. | | | LDDSE (FK.TRK) | 2.5″ × 3.5 × 16″ | 5 lb | to bars | 3 times daily | 16 ft | Fork lift |
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**What is moved?**

# Unit Load Design

Unit load: Picked up and moved between two locations as a single mass.

Smaller unit loads:

- increase transportation requirements
- reduce work in progress inventory

Larger unit loads:

- may require bigger and heavier equipment/larger aisles
- reduce the number of moves required

# Unit Load Design

Two important elements in determining the size of the unit load:

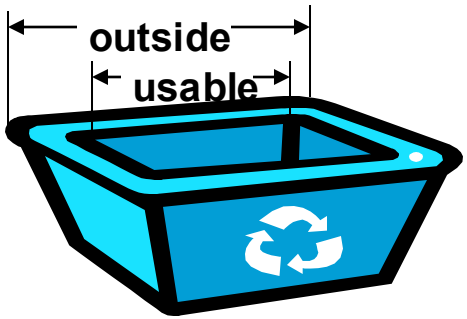
- Cube (volume) limit
- Weight limit

Four basic methods to move the unit load:

- **Lifting** under the mass
- **Inserting** the lifting element into the body of the unit load
- **Squeezing** the load between two lifting surfaces
- **Suspending** the load

Containers with good stacking and nesting features can provide significant reduction in material handling costs

# Returnable containers




Features
- Stackability
- Nestability

Efficiency of returnable containers

Container Space Utilization

= usable cube(inside) / container envelope(outside)

Container Nesting Ratio

= container height : nested height

Trailer Space Utilization

= usable storage space in trailer / inside trailer dimensions

# Returnable containers

EX1: Trailer Transports Containers **(Length x Width x Height)**

- Containers are not palletized
- No clearance between containers, or between containers and walls

Cube Inside Dimensions: 18´´ x 11´´ x 11´´
Cube Outside Dimensions: 20´´ x 12´´ x 12´´
Each Nested Container: 20´´ x 12´´ x 2´´
Inside Trailer Dimensions: 240´´ x 120´´ x 120´´
Storage Opening Dimension: 24´´ x 16´´ x 14´´

**Determine:**

- Container Space Utilization
- Container Nesting Ratio
- Trailer Space Utilization

Container Space Utilization (usable space of the unit(cube))
= usable cube(inside)/container envelope(outside)

Container Nesting Ratio (# of cubes nested that are the same size as a closed stacked cube)
= container height/ nested height

Trailer Space Utilization (Actual space in the trailer used for storage)
= space by max # containers in trailer / inside trailer dimensions

# Returnable containers

EX1: Trailer Transports Returnable Containers

- Containers are not palletized
- No clearance between containers, or between containers and walls

Cube Inside Dimensions: **18´´ x 11´´ x 11´´**

Cube Outside Dimensions: **20´´ x 12´´ x 12´´**

Each Nested Container: 20´´ x 12´´ x 2´´

Inside Trailer Dimensions: 240´´ x 120´´ x 120´´

How many can we fit? (no stack)
- 240/**20**=12 along length
- 120/**12**=10 along width
- 120/**12**=10 vertically

Total number of container is 12*10*10=1200

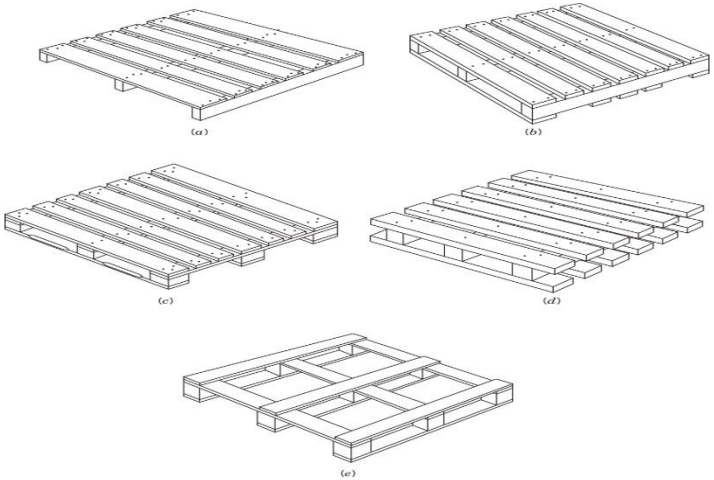
18*11*11*1200/(240*120*120)=0.76

**Determine:**

- Container Space Utilization ==========→ **2178/2880**=76%
- Container Nesting Ratio ==========→ 12:2= 6:1
- Trailer Space Utilization ==========→ 76%

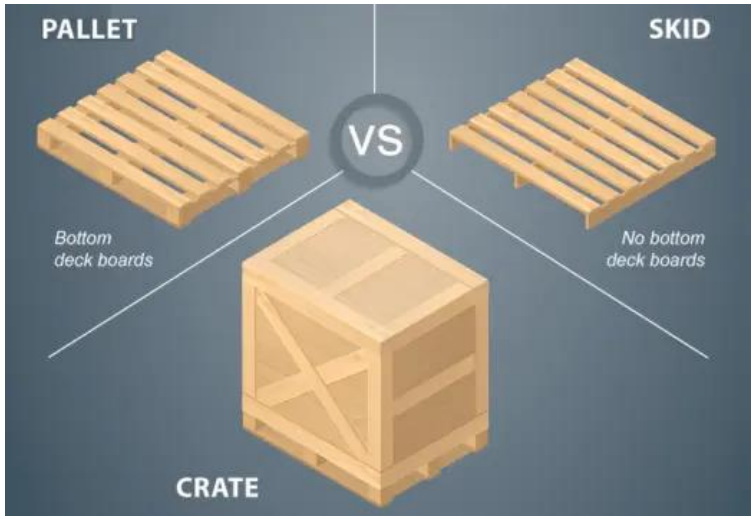
# Pallets

- Another common method of containing the unit load
- Come in a variety of designs
- Pallet loading problem: relationship between the container and pallet
- Objective is to maximize the use of space, stability for reasons of uniform loading and safety, and cost



# Material Handling Equipment

- Containers and unitizing equipment
- Containers
  - Pallets
  - Skids and skid boxes
  - Tote pans



# Material Handling Equipment

- Containers and unitizing equipment
- Unitizers
  - Stretchwrap
  - Palletizers

# Material Handling Equipment

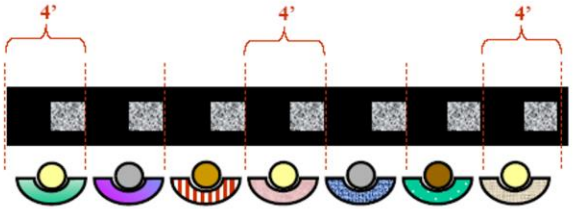
Material **Transport** Equipment: Used to move material from one location to another.

The major subcategories of transport equipment are:

- **Conveyors** –used to move materials over a fixed path between specific points
- **Cranes** –used to move materials over variable paths within a restricted area
- **Industrial Vehicles** –used to move materials over variable paths, with no restrictions on the area covered by the movement (i.e., unrestricted area)
- **No equipment** –material can also be transported manually using no equipment

# Material Handling Equipment

- Note products must be spaced 4' apart`

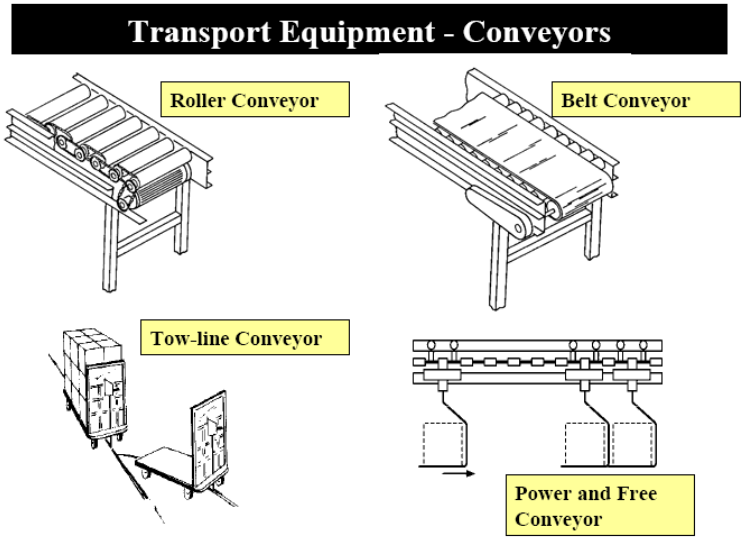



Conveyors can be classified in different ways:

- Type of product being handled
- Location of the conveyor
- Whether or not loads can accumulate (queue) on the conveyor

Types of Conveyors

- Chute
- Belt
- Roller
- Wheel
- Slat
- Chain
- Towline
- Trolley
- Power-and-free
- Cart-on-track




**Conveyor speed = Station Length/Cycle Time**

https://youtu.be/tqLYhhV7u7Y

# Material Handling Equipment

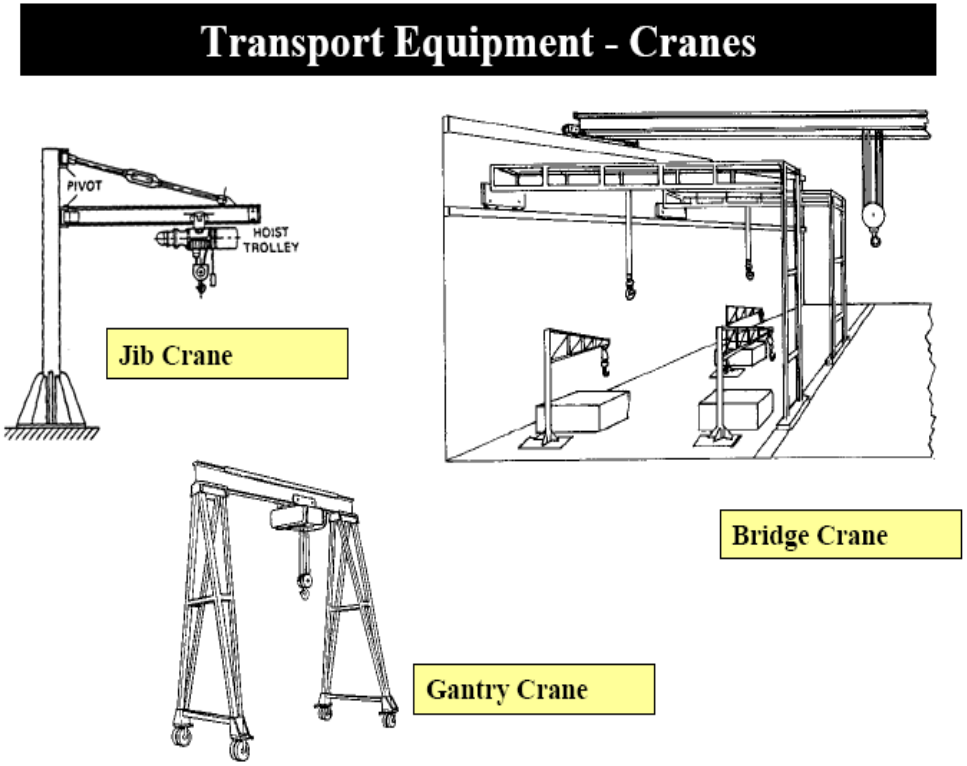
**Cranes**

- Used to move loads over variable (horizontal and vertical) paths within a restricted area
- Used when there is insufficient (or intermittent) flow volume such that the use of a conveyor cannot be justified
- Provide more flexibility in movement than conveyors
- Provide less flexibility in movement than industrial vehicles
- Loads handled are more varied with respect to their shape and weight than those handled by a conveyor

# Material Handling Equipment

Types of Cranes:

- Jib crane
- Bridge crane
- Gantry crane
- Tower crane
- Stacker crane



# Material Handling Equipment

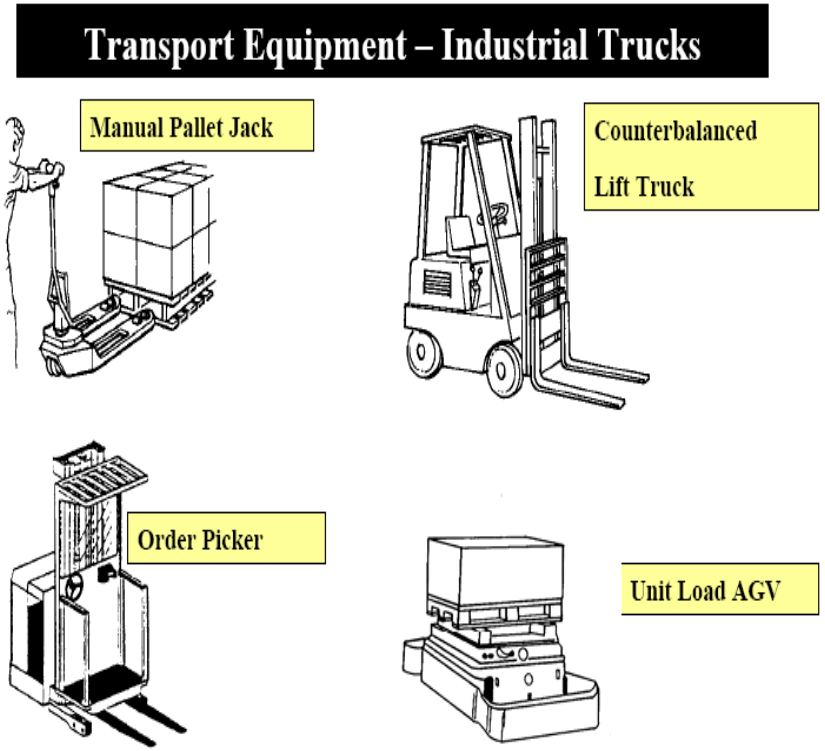
## Industrial Vehicles

- Used to move materials over variable (horizontal) paths with no restrictions on the area covered (i.e., unrestricted area)
- Provide vertical movement if the vehicle has lifting capabilities
- Used when there is insufficient or intermittent flow volume such that the use of a conveyor cannot be justified
- Provide more flexibility in movement than conveyors and cranes
- Not licensed to travel on public roads –"commercial trucks" are licensed to travel on public roads.

# Material Handling Equipment

Types of Industrial Vehicles

- Walking
  - Hand trucks and hand carts
  - Pallet jacks
  - Platform trucks
  - Walkie stackers
- Riding
  - Pallet truck
  - Tractor trailer
  - Counterbalanced lift truck
  - Straddle carrier
  - Mobile yard crane
- Automated
  - Automated guided vehicles
  - Automated electrified monorails
  - Sorting transfer vehicles



# In-class Activity

You are given the following dimensions, calculate the **Trailer Space Utilization.**

Cube Inside Dimensions: 6´´ x 4´´ x 4´´
Cube Outside Dimensions: 7´´ x 5´´ x 5´´
Each Nested Container: 7´´ x 5´´ x 1´´
Inside Trailer Dimensions: 90´´ x 60´´ x 70´´

# In-class Activity

## Answer

90/7=12 along length
60/5=12 along width
70/5=14 vertically
Total number of container is 12*12*14=2016
6*4*4*2016/(90*60*70)=**0.512 or 51.2%**

# Material Handling Equipment

**Storage equipment:**

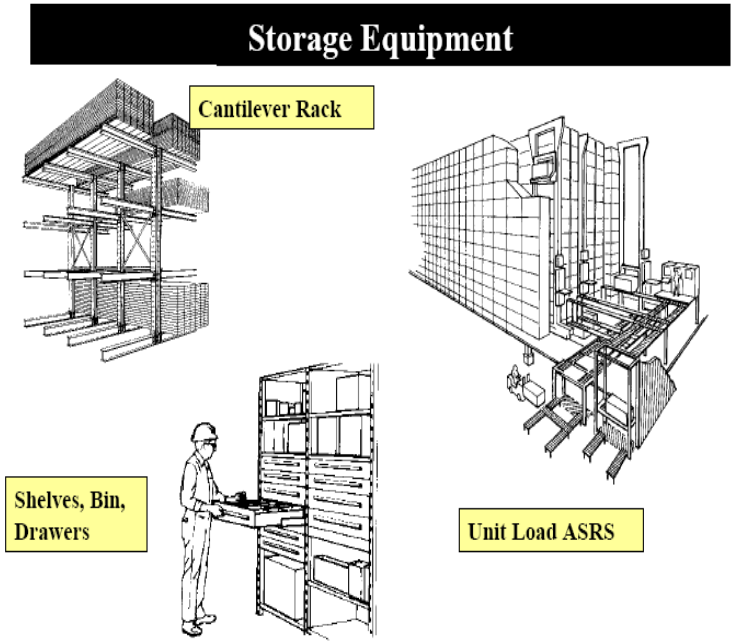
- It is used for holding or buffering materials over a period of time.
- Some storage equipment may include the transport of materials.
- If materials are block stacked directly on the floor, then no storage equipment is required.

# Material Handling Equipment

**Storage Equipment**

**The most common equipment used are**:

- Block stacking (no equipment)
- Selective pallet rack
- Drive-through rack
- Drive-in rack
- Flow-through rack
- Push-back rack
- Sliding rack
- Cantilever rack
- Stacking frame
- Shelves/bins/drawers
- Storage carousel
- Automatic storage/retrieval (AS/RS)



4 Docs
Robots run Amazon's warehouse?

# Material Handling Equipment

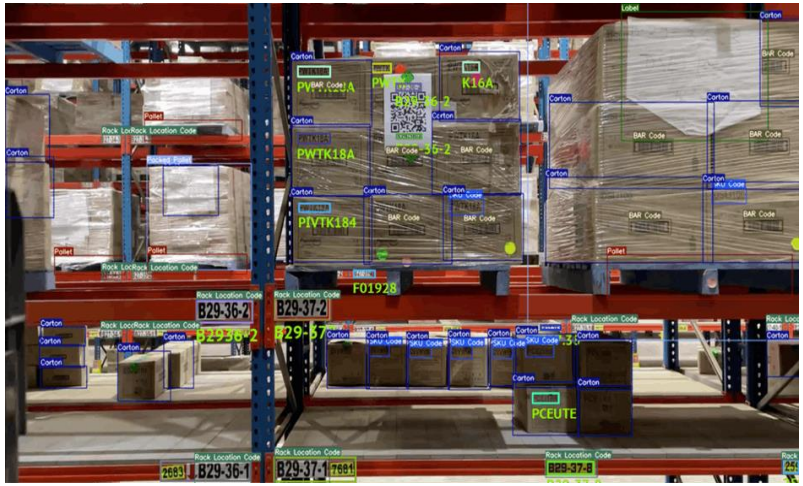
**Automatic data collection and communication equipment:**

- It is used to collect and communicate the information that is used to coordinate the flow of materials within a facility and between a facility and its suppliers and customers.
- The identification of materials and associated controls can be performed manually with no specialized equipment

# Material Handling Equipment

**Automatic data collection and communication equipment:**
The major types of identification and communication equipment are:

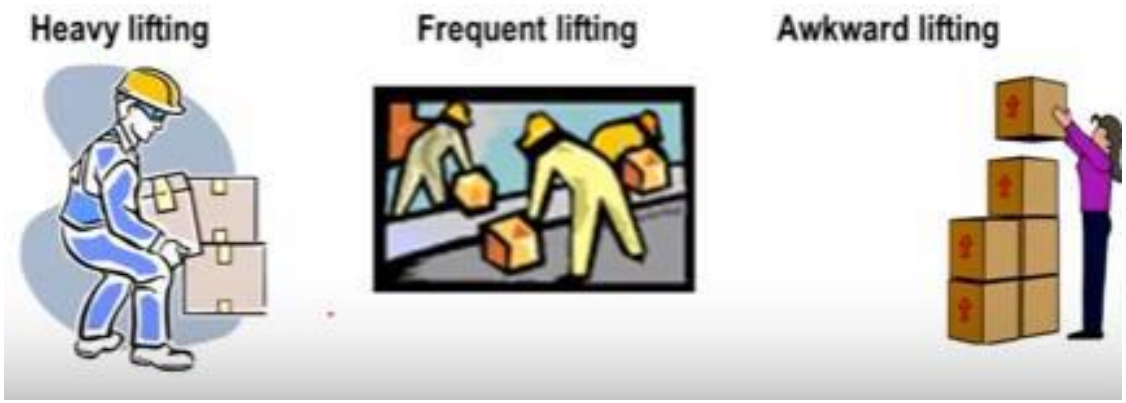
- Automatic identification and recognition
  - Bar coding
  - Optical character recognition
  - Radio frequency (RFID) tag
  - Magnetic stripe
  - Computer vision
- Automatic paperless communication
  - Radio frequency data terminal
  - Voice headset
  - Light and computer aids
  - Smart card



# Safety Considerations

- Scientific research shows an increase in injuries at certain levels of exposure to heavy, frequent, and awkward lifting

- Job design is very important in creating a safe and healthy work environment and the selection of proper material handling devices can help with reducing some of the material handling related injuries.



# Material Handling Equipment

Principles for reducing **heavy** lifting
- Reduce the weight
- Use mechanical assistance
- Slide instead of lift
- Team lifting

Principles of reducing **frequent** lifting
- Use mechanical assistance
- Avoid unnecessary lifting
- Use mobile storage

Principles for reducing **awkward** lifting–**reaching**
- Remove obstacles
- Slide closer
- Reduce shelf depth
- Adjustable storage shelves

# Material Handling Equipment

Principles for reducing **awkward** lifting–**bending**

- Use mechanical assistance
- Adjustable storage shelves
- Avoid unnecessary lifting
- Add handles

Principles for reducing **awkward** lifting–**reaching** above the shoulders

- Adjustable storage shelves
- Add handles
- Use a rolling stair or safety ladder

Principles for reducing **awkward** lifting–**twisting**

- Use conveyors
- Provide more space
- Adjustable storage shelves